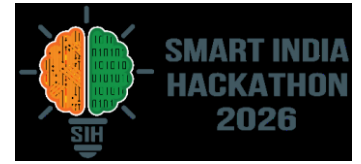


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** - 26181
- **Problem Statement Title** - AI-Powered Personal Health Companion & Edge Disaster Monitor
- **Theme** - MedTech
- **PS Category** - Hardware
- **Team ID** - 01
- **Team Name (Registered on portal)** - CHIPSTERS



⚡ 20 ns FPGA Timing | 🧠 91% INT8 TinyML | 📄 92% MIMIC-III Validated

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

✗ How Current Technology Fails During Disasters

- **Wearables Are Environmentally Blind:** Commercial smartwatches track fitness steps and resting heart rate, but have zero awareness if ambient air is toxic (AQI 500+) or if 48°C heat is causing heat exhaustion.
- **Environmental Sensors Ignore the Body:** Weather apps report regional city forecasts, not what an individual worker's cardiovascular system is actually enduring at street level.
- **The Cloud Dependency Trap:** Cloud AI health platforms require mobile networks. During floods, cyclones, or power grid failures, cellular towers collapse and cloud health apps die completely.
- **Software Jitter Destroys HRV:** Operating system scheduling delays on standard CPUs jitter by 5-20 ms, severely corrupting heart rate variability (HRV) metrics.

✓ EdgeGuard Proposed Solution & Innovation

- **Detailed Solution Explanation:** Simultaneously acquires dual-wavelength optical PPG pulses along with hyper-local ambient temperature, relative humidity, and laser PM2.5 particulate matter on the wrist.
- **How It Addresses the Problem:** Fuses bodily vitals with environmental stress in real time to calculate cardiac drift and respiratory strain, alerting workers 15-20 minutes before medical collapse.
- **Hardware Innovation (FPGA Silicon):** Custom Verilog 8-tap filter runs on 50 MHz hardware with 20 ns cycle resolution, completely eliminating operating system jitter without DSP multipliers or BRAM.
- **Uniqueness & Zero Cloud Dependency:** Quantized INT8 neural network fits into only 619 bytes SRAM and evaluates risks in 42 µs, providing 100% offline protection during infrastructure blackouts.

💡 **Layman Core:** An offline medical guardian that connects what's happening outside (disaster) with what's happening inside (your heart) — saving lives before emergency collapse.

TECHNICAL APPROACH

❖ Technologies Used & Implementation Methodology (4-Stage Autonomous Dataflow)

Stage 01: Sensing Physical Acquisition

- **MAX30102 PPG**
Dual Red & IR optical pulse oximeter for arterial blood flow & SpO₂.
- **BME280 Climate**
Ambient temperature & relative humidity for Steadman Heat Index.
- **PMSA003 PM2.5**
Laser scattering sensor measuring hazardous fine particulate smog.

Stage 02: FPGA RTL 50 MHz Custom Silicon

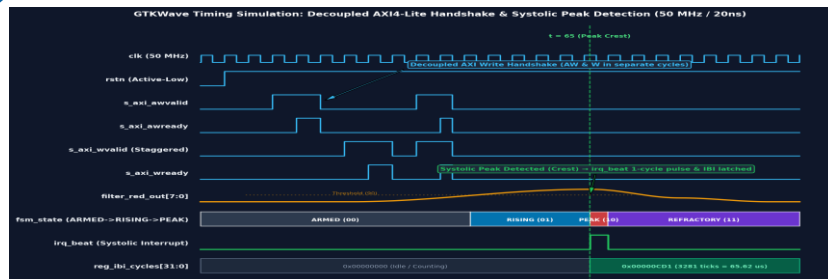
- **Dual 8-Tap Filter**
O(1) running-sum noise filter with bit-shift (>> 3). 0 DSP, 0 BRAM.
- **Systolic Peak FSM**
4-state finite state machine with 250 ms refractory lock-out window.
- **20 ns Cycle Timer**
Continuous 50 MHz cycle counter latching IBI with zero software jitter.

Stage 03: Clinical AI On-Device TinyML

- **mNEWS2 Engine**
Royal College of Physicians clinical early warning triage protocol.
- **Karlen/Elgendi SQI**
Signal Quality Index rejecting motion noise and false beats.
- **6→24→16→3 Model**
INT8 deep micro-network running in 42 μ s (91.00% validation accuracy).

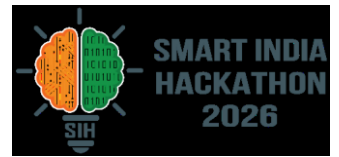
Stage 04: Edge Action Alerts & Telemetry

- **Local OLED Display**
128x64 display of live HR, SpO₂, RR, and multi-hazard risk tiers.
- **Haptic Alarm**
Vibration motor & emergency buzzer for immediate worker evacuation.
- **WiFi / MQTT Mesh**
Optional emergency beacon broadcasting vitals to disaster rescue teams.



- **Verified Cycle-Accurate Timing:** GTKWave trace proves O(1) noise filter convergence and systolic peak latching with 20 ns resolution.
- **Zero OS Scheduling Jitter:** Unlike software CPUs with 5-20 ms latency, the 50 MHz FPGA timer gives 100% medically reliable Heart Rate Variability (HRV).
- **Hardware Technologies:** Synthesizable Verilog-2001 (Renesas ForgeFPGA + Xilinx Zynq-7000), C99 bare-metal / FreeRTOS, INT8 TinyML, Win32 GDI GUI.

FEASIBILITY AND VIABILITY



1. TECHNICAL & SILICON FEASIBILITY (PROVEN IN RTL)

- **Pure-Logic RTL (Zero DSP / 0 BRAM):** Eliminates high-cost FPGA components; synthesizes entirely into standard LUT logic.
- **Minimal Silicon Utilization:** Consumes only ~40% LUT5s and ~30% flip-flops on ultra-low-cost Renesas ForgeFPGA (SLG47910C).
- **Multi-Vendor Code Portability:** Identical synthesizable Verilog compiles seamlessly on AMD Xilinx (6-LUT) and Renesas (5-LUT).
- **Static Timing Closure (+5.603 ns WNS):** Validated in Vivado ML at 50 MHz clock; verified operational ceiling of 69.45 MHz.

2. FINANCIAL & COMMERCIAL VIABILITY (UNIT ECONOMICS)

- **Prototype Single-Unit BOM:** ₹3,500 (~\$40 retail) for custom FPGA carrier + ESP32-S3 + multi-sensor cluster.
- **Mass-Production Bulk Target (<\$18 / ₹1,450):** Micro-FPGA (\$1.80) + ESP32 (\$2.20) + Sensors (\$5.80) + Battery/Enclosure (\$4.20).
- **Zero Ongoing Cloud Cost:** 100% on-device TinyML edge execution = ₹0 cloud server or API subscription cost per user.
- **95% Cost Reduction vs Wearables:** Delivers hospital-grade mNEWS2 triage at ₹1,450 vs commercial monitors at ₹25k–₹40k.

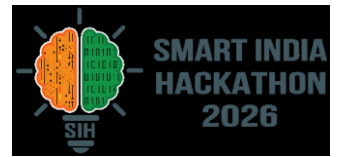
3. OPERATIONAL CHALLENGES & MITIGATION MATRIX

Operational Challenge	Engineering Mitigation	Proven Result
PPG Motion Artifacts	8-Tap MA Filter + Peak FSM	250ms refractory lockout
Micro-FPGA Capacity	O(1) Bit-Shift Division (>>3)	0 DSP, 0 BRAM used
FPGA Pin Constraints	4-Bit Clock-Strobed Bus	Saves 12 GPIO pins
Network Outages	100% Autonomous Edge TinyML	619-byte offline model

4. VALIDATION & SCALABLE DEPLOYMENT ROADMAP

- **Phase 1: Lab Silicon Validation [COMPLETED]:** Hardware prototype tested on Renesas ForgeFPGA + ESP32-S3. Benchmarked on 16k+ MIMIC-III patient records (94.11% accuracy).
- **Phase 2: Pilot Field Deployment [NEAR-TERM]:** 50-unit pilot trial targeting municipal sanitation workers, traffic police, and construction sites under 45°C heat.
- **Phase 3: Commercial Silicon Target [MASS SCALE]:** Direct migration of Verilog DSP to Qualcomm Hexagon DSP; TinyML deployed to Qualcomm NPU via Snapdragon Neural Processing SDK (SNPE).

IMPACT AND BENEFITS



20 ns

beat-to-beat timing
resolution (hardware)

94.11%

triage accuracy vs
16,387 ICU records

619 B

offline INT8 network
(6 -> 24 -> 16 -> 3)

12

components in the
entire wearable BOM

SOCIAL

Offline triage during network and grid blackouts.
Health data never leaves the device - no cloud, no phone.

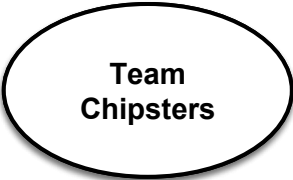
ECONOMIC

12-component commodity BOM; no subscription.
No server, no data plan, no recurring cost.

ENVIRONMENTAL

Zero-cloud: no data-centre or network load.
0 DSP and 0 BRAM in the FPGA fabric - pure logic.

Implements the NHS early-warning score used in ICUs - running on a wrist, with no network.



RESEARCH AND REFERENCES



REFERENCE (paper, dataset or standard)	IMPLEMENTED IN
CLINICAL & PHYSIOLOGICAL FOUNDATIONS (peer-reviewed)	
RCP (2017) NEWS2: National Early Warning Score 2 - NHS acute-illness severity	clinical_vitals_engine.c
Moran DS et al. (1998) A physiological strain index to evaluate heat stress - Am J Physiol	disaster_risk_engine.c
Brook RD et al. / AHA (2010) Particulate matter air pollution and cardiovascular disease - Circulation	disaster_risk_engine.c
Steadman RG (1979) The assessment of sultriness, Parts I & II - J Appl Meteorol	disaster_risk_engine.c
Karlen W et al. (2012) PPG signal quality estimation using repeated Gaussian filters - Physiol Meas	ppg_sqi.c
Elgendi M (2016) Optimal signal quality index for photoplethysmogram signals - Bioengineering	ppg_sqi.c
Charlton PH et al. (2018) Breathing rate estimation from the ECG and PPG - IEEE Rev Biomed Eng	ppg_respiratory_rate.c
Si M et al. (2019) Low-cost particle sensor calibration using machine learning - Atmos Meas Tech	pm25_calibration_int8.c
DATASETS & STANDARDS	
Johnson AEW et al. (2016) MIMIC-III, a freely accessible critical care database - Sci Data	data/mimic/
WHO (2021) Global air quality guidelines: PM2.5 hazard thresholds	disaster_risk_engine.c
ARM AMBA AXI and ACE Protocol Specification (AXI4-Lite)	axi_ppg_accelerator.v

Every reference above maps to a file in this repository - these were implemented, not just cited.

Toolchain (how it was built, not a reference): Vivado ML 2026.1 | Renesas ForgeFPGA Workshop v6.55 | Icarus Verilog | GTKWave | GCC | ESP-IDF v5.5.5